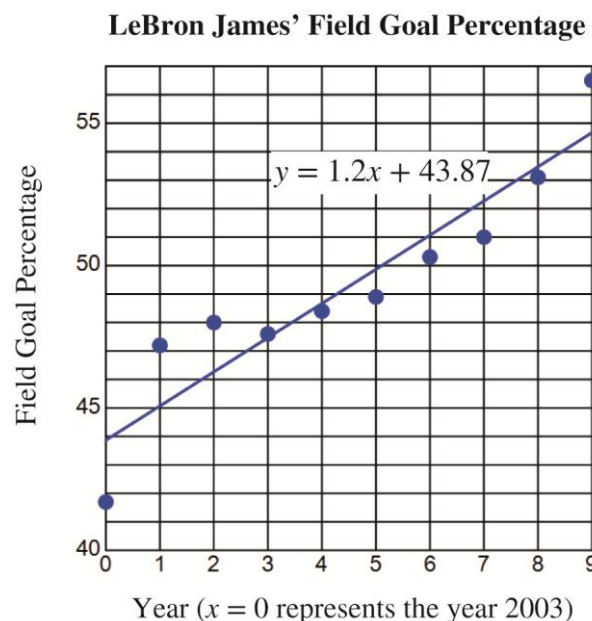


Interpreting a Linear Equation in Two Variables

Linear equations can often be used to describe (or model) the relationship between two variables in a real-world event.

1. LeBron James is one of the best all-around players in the history of the NBA. His field goal percentage, y , for each season can be approximated by the equation $y = 1.2x + 43.87$. In this equation, x represents the number of years since 2003 ($x = 0$ represents 2003, $x = 1$ represents 2004, and so on).

- a. Use the equation to approximate LeBron's field goal percentage in 2006.



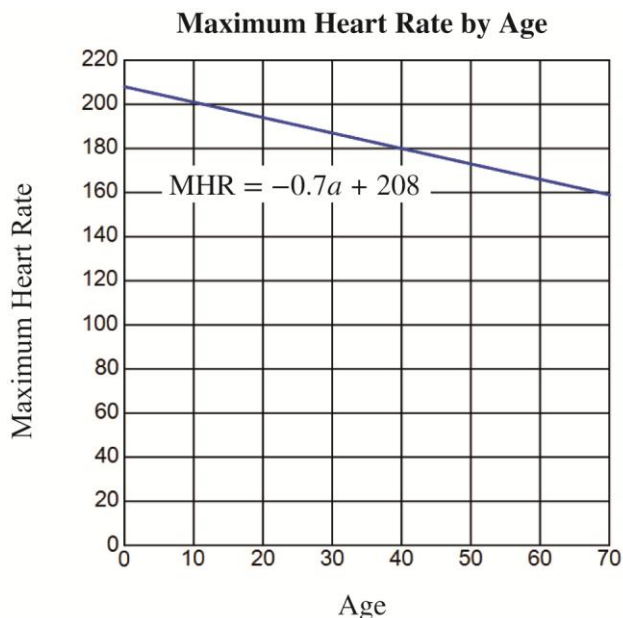
- b. Assuming this trend continues, use the equation to predict LeBron's field goal percentage in 2020.
- c. What is the slope of the line? Interpret the meaning of the slope in the context of this problem.

2. The maximum heart rate (MHR) is the maximum number of times your heart can beat in a minute during an all-out effort. The MHR depends on a person's age and can be approximated by the formula, $MHR = -0.7a + 208$. In this equation, a represents the person's age.

a. Use the equation to approximate a person's maximum heart rate if he is 20 years old.

b. Use the equation to approximate a person's maximum heart rate if he is 45 years old.

c. What is the slope of the line? Interpret the meaning of the slope in the context of this problem.

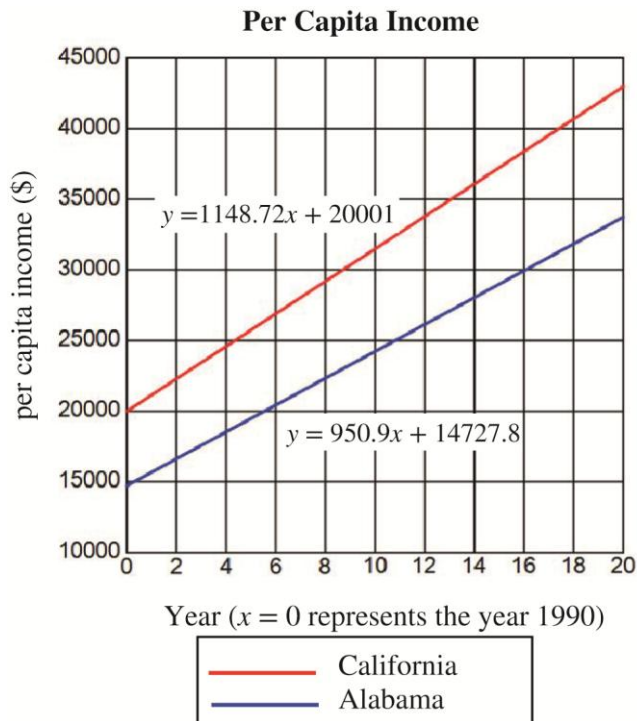


3. The graph depicts the per capita income for Alabama and California since 1990. Two linear equations are given: one to describe the per capita income for Alabama and one to describe the per capita for California. Let y represent the per capita income. Let x represent the number of years since 1990.

a. What is the slope of the line representing the per capita income of California? Interpret the meaning of the slope in the context of this problem.

b. Determine the y -intercept of the line representing the per capita income of Alabama? Interpret the meaning of the y -intercept in the context of this problem.

c. Assuming this trend continues, use the equation to predict the per capita income of Alabama in 2025.



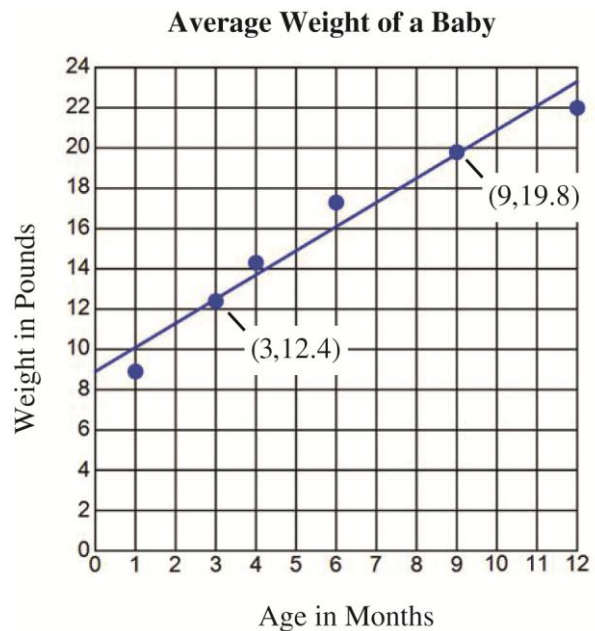
Writing a Linear Model Using Observed Data Points

4. The graph depicts the average weight for a baby by month. Let x represent the baby's age in months. Let y represent the average weight in pounds.

a. Use the points $(3, 12.4)$ and $(9, 19.8)$ to determine the slope of the line.

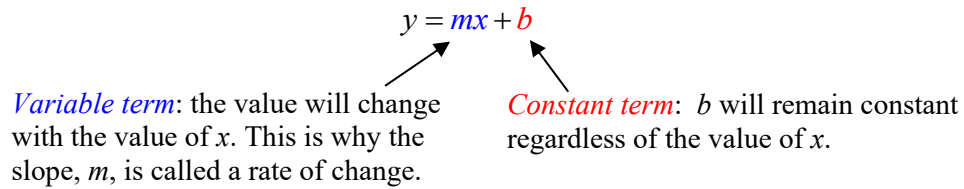
b. Interpret the slope in the context of this problem?

c. Use the points $(3, 12.4)$ and $(9, 19.8)$ to find a linear equation relating the average weight of a baby, y , to the baby's age, x .



d. If this linear trend continues beyond the observed data values, use the equation in part (c) to predict the average weight of a baby at 15 months.

Writing a Linear Model Given a Fixed Value and a Rate of Change



5. A plumber charges \$50 plus \$75 an hour to install a sprinkler system.
- Write a linear equation to compute the cost, C , for x hours of work.
 - Use the equation to compute the cost if it takes the plumber 4 hours to install the sprinkler system.

-
6. A massage therapist gets paid \$20 a day plus \$15 for each one hour massage.
- Write a linear equation to compute the total pay, y , for 1 day if x hours of massages are given.
 - Use the equation to compute the total pay for 1 day if the therapist gives 5 hours of massage.